

How the Metrics Are Computed

AI Use Impact Tracker — Technical Reference

Refined Edition

Sapien Labs | May 2026

This document is a detailed walk-through of every metric the AI Impact Tracker publishes, including the exact formulas, denominators, suppression rules, and how the dashboard's Period filter pools months together.

It is a companion to [COLUMNS.md](#) (which documents the source-side survey columns) and [TRACKER.md](#) (the spec/contract for the metric layer). All formulas match the implementation in [metrics.py](#) and [normalize.py](#).

1. Core Analytical Building Blocks

All metrics are calculated from one row per respondent per month. Three primary derived fields drive almost everything else.

1.1 AI-Use Frequency Scale (ai_freq_int)

Mapped from the raw ai_freq text answer to a 0-6 ordinal scale:

Integer	Label	Raw text mapped
0	Never	"I have never used an AI assistant"
1	Rarely	"Rarely"
2	Monthly	"A few times a month"
3	Weekly	"Several days a week"
4	Daily	"Several times a day"
5	Constantly	"Constantly", "Constamment", "Constantly (when awake)"
6	Always	"All of the time"
—	null	"N/A" or anything not recognised

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1.2 User Status (is_user)

A boolean identifying whether the respondent uses AI at all:

$$is_{user} = \mathbb{1}$$

A respondent at level 0 (“Never”) is not an AI user.

1.3 Impact Flags (from ai_impact_work)

The pipe-delimited ai_impact_work answer is split into nine boolean columns; each respondent can have multiple flags set true.

Flag	Triggered by answer
impact_improved_quality	“Improved my work quality or output”
impact_new_opportunities	“Created new job or income opportunities”
impact_adaptation_pressure	“Increased pressure to adapt or work faster”
impact_job_anxiety	“Made me worry about the future of my job or industry”
impact_job_loss	“Caused me to lose my job”
impact_reduced_income	“Reduced my income or made it harder to find work”
impact_none	“No impact”
impact_other	“Another impact not listed here” (plus any unknown raw option)
impact_not_sure	“Not sure”
impact_na	Raw value was null / N/A / blank

Two derived boolean columns roll the flags into directional vectors:

- **Positive (is_positive):** impact_improved_quality OR impact_new_opportunities
- **Negative (is_negative):** impact_adaptation_pressure OR impact_job_anxiety OR impact_job_loss OR impact_reduced_income

impact_none, impact_other, impact_not_sure, and impact_na are treated as neither positive nor negative.

2. Core Population Indicators

These base descriptive metrics are calculated independently for every demographic stratum and month block.

2.1 Total Respondent Count (N)

The absolute headcount of all individuals who completed the survey inside a designated stratum × month cell. This is also the denominator for adoption_rate.

$$N = |\{ i : i \in \text{stratum} \times \text{month} \}|$$

2.2 Adoption Rate

The proportion of the surveyed population that actively employs AI tools:

$$\text{adoption}_{\text{rate}} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\text{is_user}_i]$$

Range [0, 1]. Includes respondents at every ai_freq level from “Rarely” upward.

2.3 Frequency Distribution (freq_share₀ ... freq_share₆)

For each ordinal level $k = 0, \dots, 6$:

$$\text{freq_share}_k = \frac{1}{N} \sum \mathbb{1}[\text{ai_freq_int}_i = k]$$

The seven shares sum to 1.

3. The Impact Denominator Framework

To isolate real trends, workspace impact metrics do not use the full population headcount (N) as their base. Instead, they rely on a stricter verified sub-population called the Impact Denominator (N^d).

$$i_{\text{impact_denom}} = \text{is_user}_i \wedge \text{has_impact_response}_i$$

where has_impact_response means at least one of the eight real impact flags is true (all flags except impact_na) — i.e. the disjunction over impact_improved_quality, impact_new_opportunities, impact_adaptation_pressure, impact_job_anxiety, impact_job_loss, impact_reduced_income, impact_none, impact_other, and impact_not_sure.

To be included in the impact denominator, a respondent must fulfill two rules simultaneously:

1. **Active Engagement:** They must be a confirmed AI user ($\text{ai_freq_int} \geq 1$).

2. **Valid Response:** They must have provided an explicit answer to the impact question, including choosing baseline answers like “No impact” or “Not sure.”

Methodological Purpose: Non-users are removed from impact calculations because a participant with zero regular interaction cannot report direct workspace shifts. This step ensures that the resulting percentages accurately reflect the experiences of the user base without downward dilution.

4. Impact Share Metrics

For each impact flag listed in §1, the share is computed against the impact denominator. Let N^d denote $n_impact_denominator$:

$$impact_share_f = \frac{1}{N^d} \sum \mathbb{1}[flag(f, i)]$$

Metric	Numerator flag
impact_share_improved_quality	impact_improved_quality
impact_share_new_opportunities	impact_new_opportunities
impact_share_adaptation_pressure	impact_adaptation_pressure
impact_share_job_anxiety	impact_job_anxiety
impact_share_job_loss	impact_job_loss
impact_share_reduced_income	impact_reduced_income
impact_share_none	impact_none
impact_share_other	impact_other
impact_share_not_sure	impact_not_sure

Each is a value in $[0, 1]$.

The shares do not have to sum to 1. Respondents can select more than one impact, so a given respondent can contribute to multiple shares.

5. Positive / Negative / Net Impact

Two rolled-up shares are computed on the same impact denominator:

$$positive_impact_{share} = \frac{1}{N^d} \sum \mathbb{1}[is_positive_i]$$

$$negative_impact_{share} = \frac{1}{N^d} \sum \mathbb{1}[is_negative_i]$$

These two shares serve as building blocks for the dose-response panel (§7), which displays their difference (positive – negative) at each AI-use frequency level.

6. Weighted Impact Index (WII)

The **Weighted Impact Index** (weighted_impact_index) is the dashboard's headline metric. Unlike the simple net index in §5, it assigns each impact flag a signed weight reflecting how positive or negative that outcome is, then averages a per-respondent score across the impact denominator.

6.1 Weight Table

These are editorial values set by the Sapien Labs team; they reflect judgment about relative severity, not an empirical calibration.

Workplace Impact Choice Flag	Assigned Severity Weight
Created new job or income opportunities	+1.00
Improved my work quality or output	+0.50
Made me worry about the future of my job or industry	-0.25
Increased pressure to adapt or work faster	-0.50
Reduced my income or made it harder to find work	-0.75
Caused me to lose my job	-1.00
No impact / Another impact / Not sure	0.00

6.2 Per-Respondent Score

For each respondent i in the impact denominator, sum the weights of every flag they selected:

$$s_i = \sum_{f \in \mathcal{F}} w_e \cdot \mathbb{1}[\text{flag}(f, i)]$$

where $w_e = \text{IMPACT_WEIGHTS}[f]$ and $\mathcal{F}$ is the set of eight weighted flags (excluding impact_na).

Examples:

- A respondent who selected only “Improved my work quality” scores +0.5.
- A respondent who selected “Improved my work quality” and “Adaptation pressure” scores +0.5 + (−0.5) = 0.
- A respondent who selected “Job loss” and “Reduced income” scores −1.0 + (−0.75) = −1.75.
- A respondent who selected “No impact” scores 0.

A single respondent's score can exceed ± 1 if they selected several weighted flags in the same direction.

6.3 Cell Value

The published weighted_impact_index is the arithmetic mean of per-respondent scores over the impact denominator:

$$WII = \frac{1}{N^d} \sum s_i$$

Range is roughly $[-1, +1]$ in practice but is not symmetric: a respondent would need to select multiple negative flags simultaneously to drive a cell well below -1 .

A positive cell value means the average AI user in that stratum had a net-positive experience; a negative value means the average user had a net-negative experience.

6.4 Asymmetry by Design

The weight scheme is deliberately asymmetric: the most severe negative outcome (job loss, -1.0) carries the same absolute weight as the strongest positive (new opportunities, $+1.0$), but the cumulative negative weight across all four negative flags (-2.5) exceeds the cumulative positive weight ($+1.5$). This reflects the stakeholder judgment that severe harms deserve disproportionate weight compared to moderate benefits.

6.5 Comparison with the v1 Net Impact Index

The v1 dashboard used a simpler binary measure ($NII = \text{positive_share} - \text{negative_share}$) that treated all impacts equally. The WII replaced it as the headline metric because it differentiates by severity. In practice, global WII values (~ 0.14 – 0.18) run lower than NII values (~ 0.20 – 0.26) because the negative weights are heavier. The NII is still computed internally and drives the dose-response panel (§7).

7. Dose-Response

The dashboard's "Dose-response" row shows the difference between positive and negative impact shares (positive_impact_share – negative_impact_share) at each AI-use frequency level. For each stratum × month, and for each ordinal ai_freq level $k = 1, \dots, 6$:

$$dose_response[k] = \frac{pos_share_k - neg_share_k}{n_k}$$

where the denominator is respondents in the impact denominator (§3) at that specific frequency level. If that level has fewer than MIN_N respondents (§9), the entry is null (rendered as n/a in the dashboard).

Level 0 ("Never") is not part of the impact denominator and is not reported.

Interpretation: a dose-response that climbs from +0.10 at "Rarely" to +0.40 at "Daily" suggests the net impact is more positive among heavier users — a hint of dose-dependence, not a causal claim.

8. Confidence Intervals

8.1 Share Metrics — Wilson 95% Score Interval

All share metrics (adoption_rate, every impact_share_*, positive_impact_share, negative_impact_share) get a Wilson score 95% CI. Given k successes out of n trials and $z = 1.96$:

$$\hat{p} = \frac{k}{n}$$

$$\tilde{p} = \frac{\hat{p} + z^2/2n}{1 + z^2/n}$$

$$CI = \tilde{p} \pm \frac{z \cdot \sqrt{\tilde{p}(1-\tilde{p})/n + z^2/4n^2}}{1 + z^2/n}$$

$$ci_{low} = \max(0, CI_{low}), ci_{high} = \min(1, CI_{high})$$

Wilson is preferred over the normal-approximation Wald interval because it stays inside $[0, 1]$ even at extreme proportions and small n .

8.2 Weighted Impact Index — Wald 95% CI

The per-respondent score s_i (§6) is treated as a continuous variable:

$$SE = \frac{\sigma}{\sqrt{N^d}} (ddof = 1)$$

$$CI = WII \pm 1.96 \times SE$$

where σ_s is the sample standard deviation of the per-respondent scores and N^d is the impact denominator size.

All CIs assume simple random sampling — they will be revised once survey weights are introduced.

9. Minimum-N Suppression

Any stratum $\times$ month cell with fewer than **MIN_N (default 50)** respondents is flagged suppressed = true and its metric values are written as null. The dashboard does not display suppressed cells.

The threshold applies at the finest stratum level. So if a country $\times$ gender $\times$ age cell has 41 respondents, that cell is suppressed, even if the corresponding country $\times$ gender (without age) cell has 600.

MIN_N is configurable per run via the --min-n flag to main.py. The active value is stored in _meta.json and embedded in the rendered HTML as the suppression threshold shown in the dashboard subtitle.

The dashboard also applies a separate UI threshold: the Gender and Age band dropdowns only list categories that have at least one non-suppressed cell — so options with no usable data anywhere are hidden.

10. Stratification (Country $\times$ Gender $\times$ Age)

Every metric is computed independently for each of these eight stratum levels:

Level	Group-by columns (+ year, month)
global	—
country	country_clean
gender	gender_clean
age_band	age_band
country_gender	country_clean, gender_clean
country_age_band	country_clean, age_band
gender_age_band	gender_clean, age_band
country_gender_age_band	country_clean, gender_clean, age_band

The dashboard picks the finest level needed for the current filter combination so every displayed number is a precomputed cell value, not a JS-side aggregation:

Filters	Level used
no demographic filter	country
gender only	country_gender
age only	country_age_band
gender + age	country_gender_age_band

This is also why removing a demographic filter doesn't just average the cells back together — the dashboard re-loads the coarser stratum.

11. Rolling-Window Pooling (Period Filter)

The dashboard's Period selector — Single month / Last 3 / Last 6 / Last 12 — pools the precomputed monthly cells into a rolling window. Pooling is done in the browser on the embedded JSON and does not change the published Parquet output.

Field	Aggregation	Weight
n_respondents, n_impact_denominator	sum	—
adoption_rate	weighted mean	n_respondents
weighted_impact_index, net_impact_index, all impact_share_*, positive/negative share	weighted mean	n_impact_denominator
dose_response[k] (k = 1...6)	weighted mean	n_impact_denominator

This is approximate. The exact pooled value would require re-running the metric layer on the pooled respondent-level data. The dashboard's weighted-mean of monthly aggregates is correct in expectation but ignores within-month variance and does not produce new confidence intervals.

Cells suppressed in a given month (originally below MIN_N) are dropped before pooling, so a country can appear in the rolling window even if some of its constituent months were individually suppressed.

11.1 Pooling Formula

$$x_{pooled} = \frac{\sum n_m \cdot x_m}{\sum n_m}$$

where m indexes months in the window and n_m is the appropriate weight column for the metric class.

12. Summary Card (Dashboard)

The top-right summary card on the dashboard is a fixed reference and is not affected by Month, Period, Country, Gender, or Age filters. It always shows:

3. The global weighted average of the currently-selected metric over the trailing 3 full months.
4. The total respondents over those 3 months.
5. The total respondents across the entire dataset to date.

“Full” means non-partial. The dashboard detects the latest partial month by comparing each month's total respondents against the median of prior months — any month with respondents below $60\% \times \text{median}$ is considered still in-flight and excluded from the trailing-3 window. This is the same rule used to mark the partial point in the trend chart.

Only the metric dropdown updates this card; switching metric swaps the displayed label and value but does not change the underlying 3-month window.

13. Known Limitations

These limitations apply to every metric above.

13.1 No Survey Weights (Non-Probability Sample)

The GMP respondent pool is a convenience sample recruited via online ads (Google Display, Meta, and organic search). It carries no post-stratification or design weights. Demographic proportions in the data do not match national population proportions — countries, age groups, and genders are represented according to who encounters and completes the survey, not according to census benchmarks.

All metrics are descriptive statistics of the responding population, not population-level estimates. Associations between `ai_freq` and impact outcomes are correlational; higher-frequency AI users who report better outcomes may differ systematically from lower-frequency users in education, occupation, digital literacy, or other unmeasured confounders. No causal claims are supported.

Planned mitigation: Demographic-weighted averaging using UN World Population Prospects age–sex distributions is on the roadmap. This would rebalance age and gender composition within each country but would not address self-selection or unmeasured confounders.

13.2 Composition Effects in Time-Series

Month-over-month changes may reflect shifts in who is responding rather than genuine attitudinal change. If a recruitment campaign increases responses from a younger demographic in one month, and younger respondents tend to report more positive AI impacts, the global WII will rise even if no individual changed their view. Country mix changes and seasonal patterns in survey completion can similarly shift metrics.

The stratified breakdowns (by country, gender, age) partially mitigate this by holding one dimension constant, but cross-dimensional composition effects remain. Rolling-window views help smooth short-term fluctuations.

13.3 Weight Selection

The nine IMPACT_WEIGHTS (§6) were chosen by stakeholder judgment, not derived from empirical data. Different weight schemes would produce different index values and could alter relative country rankings. Sensitivity analysis (varying weights within a plausible range) has not yet been conducted. The asymmetric design — total negative weight (−2.5) exceeding total positive weight (+1.5) — is intentional but should be disclosed when comparing WII values across studies.

13.4 Rolling-Window Approximation

Rolling-window pooling (§11) uses total $n_{\text{respondents}}$ as approximate weights. The exact calculation would re-aggregate from individual-level data within the pooled window. The approximation introduces minor error (± 1 percentage point within 3–6 month windows) because the impact denominator differs slightly from total N . This trade-off is accepted for client-side performance.

13.5 Data Source Filtering Not Yet Implemented

Per Tara's specification, the production version should restrict the respondent pool to Google Display and Meta traffic sources and down-weight organic/search traffic to 10%. This is not yet implemented; the current pipeline processes all traffic sources equally.

13.6 Multi-Select Interaction Effects

The WII treats each flag independently — a respondent selecting both “improved quality” (+0.5) and “job anxiety” (−0.25) receives the sum (+0.25). This additive model does not capture potential interactions between simultaneous positive and negative experiences.

13.7 Suppression and Small-Cell Noise

Strata with fewer than 50 respondents (MIN_N) are suppressed entirely. For strata just above the threshold (e.g., $N = 55$), confidence intervals are wide and point estimates can shift substantially with small data changes.

13.8 CIs Assume Simple Random Sampling

All confidence intervals are computed under a simple-random-sampling assumption. The GMP is a non-probability convenience sample, so the intervals represent variability due to finite sample size

but do not account for design effects, clustering, or non-response bias. Confidence intervals will be revised once survey weights are introduced.